Nguyen Anh Tuan

Full-Stack Engineer · TypeScript / Node.js
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Education

INSA Centre Val de Loire — Bourges
Engineering degree, Security and Information Technology — *2022 - 2026*
Selective Franco-Vietnamese integrated preparatory programme, Hué — *2020 - 2022*

Experience

Snowpack (ZTNA) — Paris
Full-Stack Engineer (fixed-term contract) — *05/2026 - Present*

Team of two developers working directly with the CTO: I take each feature from spec to production on my own.

- Production incident: found the cause of recurring CPU spikes by tracing them in Grafana, then rewrote the scheduling of our automatic tasks so a run never overlaps the previous one, and capped incoming reports at **2 MB**. **CPU** from **100%** at peak to **10-20%** at most.
- Production monitoring in Grafana, fed by the audit log I built: sign-ins (success and failure) and company actions.
- Node.js upgrade (18 to 24): pre-compiled the backend. Start-up **2.5 to 1.5 min**, every CI pipeline **1-1.5 min** faster, deployed image **30%** smaller, 5 key API endpoints **1.5x to about 9x** faster.
- Self-hostable console, for a client to set up the whole Snowpack infrastructure offline: bundled the API into a single binary (**Node SEA**), packaged all of it as a **Docker Compose** stack with a **DHCP** server, and integrated it with our desktop, CLI and router (Snowgate) apps. Tested on **AMD64** and **ARM**. Wrote the install guide: our devs set up the **internal testing pipeline** from it alone.
- **Stack**: TypeScript, Node.js, ESM, swc, esbuild, Node SEA, Docker Compose, k6, Grafana, Vitest, GitLab CI.

Full-Stack Developer (internship) — *02/2025 - 03/2026*

- Group-based access, full ownership from spec to implementation: **bulk assignment** of users and devices to services, **access resolution** and **licence distribution** across groups. First release in under **2 months**, with **42 end-to-end scenarios** and **48 role-permission checks**.
- Identity providers (**Keycloak**, **Microsoft Entra ID**): each company's groups and members kept up to date with its identity provider, every 5 minutes and at sign-in. Fixed an **SSRF** vulnerability (blocked internal addresses).
- Faster data loading on every console page: users page **10-15 s down to 4-5 s** (**2-3x faster**), by replacing a shared "reload everything" with per-page loading and a short cache.
- **Data usage tracking**: daily usage per user, device and partner from the raw logs, with admin charts.
- **Stack**: TypeScript, Node.js, TSOA / OpenAPI, React, MongoDB, TypeORM, Keycloak, Docker, GitLab CI.

Projects

RFP proposal scorer (24h hackathon, team of 4) — Frankfurt
Architecture lead, Python (FastAPI), React · github.com/UNtergas/SiviHack-2026-Team — *09/2026*

- AI reviewer for sales proposals: rules in code for amounts and dates; the AI model (Gemini Flash) scores **7** criteria and must quote the proposal; code checks each quote exists.
- Tested on **10** real public bids (Michigan): matched the State's ranking on **2 of 3** contracts, under **$0.60** per bid.
- Pipeline: RFP extracted once (cached) → **3** parallel scoring groups, JSON schema → weighted score in code.

Appius, city-builder game (team of 6) — INSA Centre Val de Loire
School project, Python · github.com/jumscrafteur/appius — *10/2022 - 01/2023*

- Split the tasks and gave everyone time to contribute. Three weeks before the deadline, took the rest in hand and wrote **about three quarters of the code**.
- **Core engine of the main game**: in-game events (fire, housing, walkers, patrols, water), **A* pathfinding**.
- Rendering: **map pre-rendered once, only changed tiles redrawn** (frame rate), 4 draw layers, minimap.

Technical Skills

Programming languages: TypeScript, JavaScript, Python, SQL
Backend: Node.js, Express, TSOA, FastAPI, REST, OpenAPI / Swagger, MongoDB, TypeORM
Frontend: React, Next.js, Vite
Security & Identity: authentication, RBAC, application security, Keycloak, Microsoft Entra ID
Tools & DevOps: Git, Linux, Bash, Docker, Docker Compose, GitLab CI, Grafana, Node SEA
Testing & practices: Vitest (TDD, end-to-end), k6, Agile, object-oriented programming (OOP), code review
Spoken languages: English C1 (TOEIC 970/990), French B2 (DELF B2, 72.5/100), Vietnamese native